

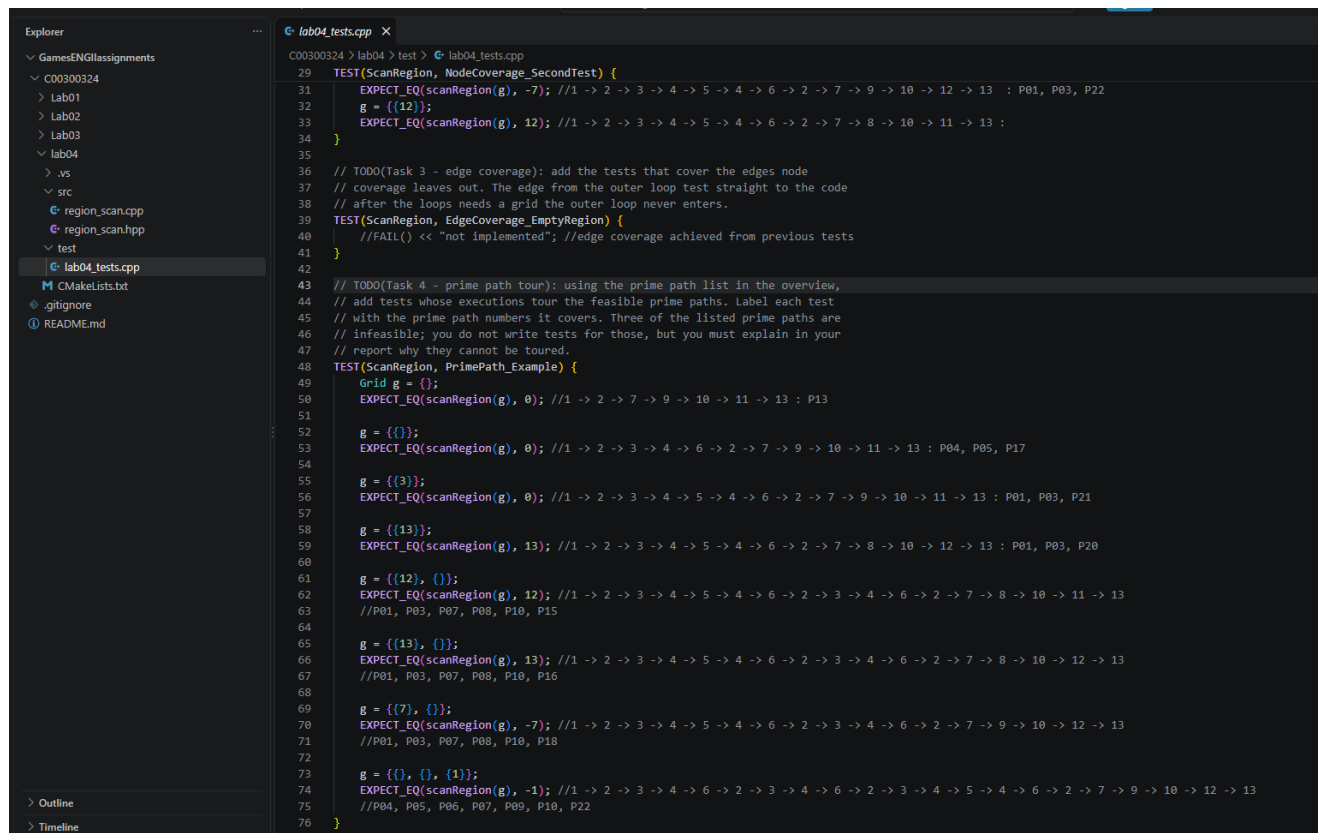
Lab 04 Report + Screen Shots

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This lab highlighted the importance of covering all tests and how it should be done with as little tests as possible, for example in this lab, ensuring node coverage will also ensure full edge coverage making it so you do not need to create extra tests for edge coverage although this may not apply for all graphs.

It is also best to create tests to cover your prime paths by going from the largest to smallest as this will result in the least number of tests as covering the larger ones will often contain the smaller prime paths inside of them allowing you to cover all prime paths efficiently.

It is best to achieve all coverage by using the least amount of tests to reduce overhead cost.



```
29 TEST(ScanRegion, NodeCoverage_SecondTest) {
30     EXPECT_EQ(scanRegion(g), -7); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 7 -> 9 -> 10 -> 12 -> 13 : P01, P03, P22
31     g = {{12}};
32     EXPECT_EQ(scanRegion(g), 12); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 7 -> 8 -> 10 -> 11 -> 13 :
33 }
34
35 // TODO(Task 3 - edge coverage): add the tests that cover the edges node
36 // coverage leaves out. The edge from the outer loop test straight to the code
37 // after the loops needs a grid the outer loop never enters.
38 TEST(ScanRegion, EdgeCoverage_EmptyRegion) {
39     //FAIL() << "not implemented"; //edge coverage achieved from previous tests
40 }
41
42 // TODO(Task 4 - prime path tour): using the prime path list in the overview,
43 // add tests whose executions tour the feasible prime paths. Label each test
44 // with the prime path numbers it covers. Three of the listed prime paths are
45 // infeasible; you do not write tests for those, but you must explain in your
46 // report why they cannot be toured.
47 TEST(ScanRegion, PrimePath_Example) {
48     Grid g = {};
49     EXPECT_EQ(scanRegion(g), 0); //1 -> 2 -> 7 -> 9 -> 10 -> 11 -> 13 : P13
50
51     g = {{}};
52     EXPECT_EQ(scanRegion(g), 0); //1 -> 2 -> 3 -> 4 -> 6 -> 2 -> 7 -> 9 -> 10 -> 11 -> 13 : P04, P05, P17
53
54     g = {{3}};
55     EXPECT_EQ(scanRegion(g), 0); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 7 -> 9 -> 10 -> 11 -> 13 : P01, P03, P21
56
57     g = {{13}};
58     EXPECT_EQ(scanRegion(g), 13); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 7 -> 8 -> 10 -> 12 -> 13 : P01, P03, P20
59
60     g = {{12}, {}};
61     EXPECT_EQ(scanRegion(g), 12); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 3 -> 4 -> 6 -> 2 -> 7 -> 8 -> 10 -> 11 -> 13
62     //P01, P03, P07, P08, P10, P15
63
64     g = {{13}, {}};
65     EXPECT_EQ(scanRegion(g), 13); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 3 -> 4 -> 6 -> 2 -> 7 -> 8 -> 10 -> 12 -> 13
66     //P01, P03, P07, P08, P10, P16
67
68     g = {{7}, {}};
69     EXPECT_EQ(scanRegion(g), -7); //1 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 3 -> 4 -> 6 -> 2 -> 7 -> 9 -> 10 -> 12 -> 13
70     //P01, P03, P07, P08, P10, P18
71
72     g = {{}, {1}};
73     EXPECT_EQ(scanRegion(g), -1); //1 -> 2 -> 3 -> 4 -> 6 -> 2 -> 3 -> 4 -> 6 -> 2 -> 3 -> 4 -> 5 -> 4 -> 6 -> 2 -> 7 -> 9 -> 10 -> 12 -> 13
74     //P04, P05, P06, P07, P09, P10, P22
75 }
76 }
```

```

PS C:\Users\gameuser\Downloads\lab04> .\build\lab04_tests.exe --gtest_color=yes

C:/Users/gameuser/Downloads/lab04/test/lab04_tests.cpp:68: Failure
Expected equality of these values:
  scanRegion(g)
    Which is: -7
  12

[ FAILED ] ScanRegion.PrimePath_Example (4 ms)
[-----] 4 tests from ScanRegion (10 ms total)

[-----] Global test environment tear-down
[=====] 4 tests from 1 test suite ran. (14 ms total)
[ PASSED ] 3 tests.
[ FAILED ] 1 test, listed below:
[ FAILED ] ScanRegion.PrimePath_Example

1 FAILED TEST
PS C:\Users\gameuser\Downloads\lab04> cmake --build build --target lab04_tests
[2/2] Linking CXX executable lab04_tests.exe
PS C:\Users\gameuser\Downloads\lab04> .\build\lab04_tests.exe --gtest_color=yes
Running main() from C:/Users/gameuser/Downloads/lab04/build/_deps/googletest-src/googletest/src/gtest_main.cc
[=====] Running 4 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 4 tests from ScanRegion
[ RUN ] ScanRegion.WorkedExample_PositiveMultipleOfThree
[ OK ] ScanRegion.WorkedExample_PositiveMultipleOfThree (0 ms)
[ RUN ] ScanRegion.NodeCoverage_SecondTest
[ OK ] ScanRegion.NodeCoverage_SecondTest (0 ms)
[ RUN ] ScanRegion.EdgeCoverage_EmptyRegion
[ OK ] ScanRegion.EdgeCoverage_EmptyRegion (0 ms)
[ RUN ] ScanRegion.PrimePath_Example
[ OK ] ScanRegion.PrimePath_Example (0 ms)
[-----] 4 tests from ScanRegion (7 ms total)

[-----] Global test environment tear-down
[=====] 4 tests from 1 test suite ran. (12 ms total)
[ PASSED ] 4 tests.
PS C:\Users\gameuser\Downloads\lab04>

```